ARMSTRONG
KINETIC SOLUTIONS

Supplying the world with an abundance of clean energy.

Every spring stores energy and gives it straight back. **The Armstrong Gate** is a mechanical full-wave rectifier that keeps it, starting with the suspension spring of every vehicle on the road. **U.S. Patent 12,409,695 B2**, granted September 9, 2025.

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Our vision

Energy is not something you generate. It is already there. You just need to stop giving it back.

Clean energy is usually framed as a supply problem: build more. We see an abundance problem. Useful motion is thrown away as heat every second, in places engineering has learned not to look. Our mission is to find those places and build simple, rugged machines that keep the energy, in service of a world that needs every clean watt it can get.

1 The problem: energy lost where springs oscillate

A spring is a perfect reservoir with a leak at both ends. It takes energy in, then hands it right back, and the damper next to it burns the remainder as heat.

Vehicle suspensions are the largest example. Every bump, ripple, and engine or tire vibration pumps energy into the coil springs of more than a billion vehicles, and none of it is kept. The same story repeats in rail bogies, truck cabs and seats, robot joints, ship mounts, industrial machinery, and seismic isolators.

The measurement gap. Decades of "regenerative suspension" research measured the **damper** and reported small numbers. The **spring** stroke, which carries the bulk of the stored energy, was never the target. That blind spot is the opportunity.

Road vehicles · Heavy trucks & fleets · Military vehicles · Robotics joints · Rail · Marine · Industrial vibration · Seismic

2 Why hasn't it been done?

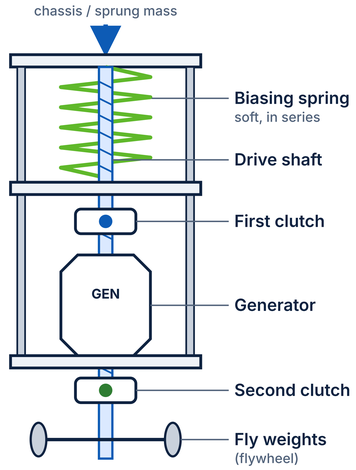
Well-funded teams have tried to pull power from suspensions. Every one of them either targeted the damper or used powered actuators that consume energy.

Bose (2004–2017)	Electromagnetic active suspension; drew 1–4 kW per corner. Never shipped; technology sold to ClearMotion.
Audi eROT (2016)	Rotary damper harvester. Measured 100–150 W average, 3 W on smooth road. Never shipped.
ZF / Levant GenShock	Hydraulic damper harvester. Pivoted away from harvesting toward ride comfort.
ClearMotion (shipping)	$365M raised, in the Nio ET9 today. Proves the market exists, but it *consumes* power to smooth the ride.

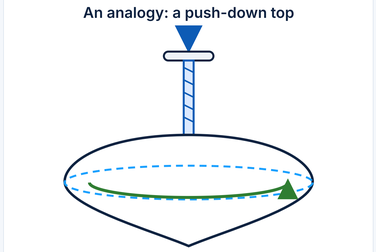
Our difference: we tap the spring stroke, not the damper, with a purely mechanical one-way drive. No actuators, no control power. And because the Gate pulls energy out of the suspension's motion, harvesting and ride quality move in the same direction rather than trading off.

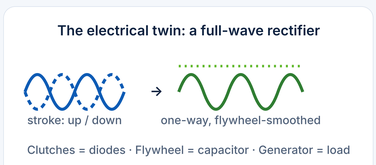
3 Our first solution: the Armstrong Gate

A gate lets traffic through in one direction. The Armstrong Gate lets spring motion through in one direction, as rotation, into a generator.



The patented drive. Both halves of the stroke turn the generator shaft the same way.





How it works

1. **A soft spring goes first.** A frame biasing spring, softer than the main suspension spring, sits in series with it. Because it is the softest element, it takes the first inches of every stroke and routes that motion into the drive before the main spring deflects.
2. **Stroke becomes rotation.** The drive shaft plunges through two opposed one-way (sprag) clutches with opposite-hand screw engagement. Compression grips the first clutch; rebound grips the second.
3. **Both directions spin one way.** Each clutch drives the hollow generator shaft in the same direction. Up and down motion becomes continuous rotation: mechanical full-wave rectification.
4. **Fly weights smooth it, the generator keeps it.** The flywheel carries rotation through every stroke reversal and the generator turns smooth rotation into electricity for the vehicle's battery. No actuators, no control power.

4 What we are aiming for

Prove it on a bench, then demonstrate it on a real vehicle with coil springs.

DONE **Patent granted**
U.S. 12,409,695 B2, 18 claims, September 2025. First prototype frame machined (pictured below).

NOW **Validated digital twin**
Event-driven quarter-car plus Gate model with an exact energy ledger. Checked against published motion-rectifier and Audi eROT data before any prediction is reported.

NEXT **Bench test at Duke**
Shaker-rig stroke sweeps at 3–5 pre-registered amplitude and frequency points, with pass/fail bands written down before the test runs.

GOAL **On-vehicle demonstration**
Install the Gate on a coil-spring vehicle (a full-size pickup is the target) and measure harvested power across real roads, so the concept is demonstrated, not argued.

5 The story



Larry Armstrong, inventor, of Eufaula, Oklahoma, spent a career building machines and asked a question the industry had stopped asking: why does every spring give its energy back? The answer became the Armstrong Gate.

Avery Armstrong, Duke University, co-founded Armstrong Kinetic Solutions to carry the patent from drawings to hardware, with Duke's labs as the proving ground.

Pictured: Avery and Larry with the first machined Gate frame, summer 2025.

We are a family company building for the whole world. Come find us at the poster; we would love your questions, your skepticism, and your help.

6 Honest numbers

QUANTITY	VALUE	BASIS
Motion-rectifier mechanical efficiency (prior art)	**> 60 %**	LITERATURE Li & Zuo, 2013
Damper-only harvest, Audi eROT	**100–150 W avg**	LITERATURE Audi, 2016
Passenger car, spring-stroke harvest	**hundreds of W**	PREDICTED Gate twin, pre-prototype
Heavy trucks, military, rough duty	**kW class**	PREDICTED road input is the limit
Regenerative suspension market	**$0.64B → $10B**	ESTIMATE 2025 → 2035

LITERATURE published measurement · PREDICTED our model, not yet measured · ESTIMATE third-party market figure

A sedan will not charge itself. The large numbers live where road energy input is large: heavy vehicles, fleets, off-road and defense.